



Boundary optimization for rough sets

Konrad Engel^{a,*}, Tran Dan Thu^b

^a University of Rostock, Institute for Mathematics, Ulmenstraße 69, 18057 Rostock, Germany

^b School of Information Technology, University of Science Ho Chi Minh City, 227 Nguyen Van Cu, Dist. 5, Ho Chi Minh City, Viet Nam



ARTICLE INFO

Article history:

Received 8 December 2017

Received in revised form 24 May 2018

Accepted 26 May 2018

Available online 14 June 2018

Keywords:

Rough sets

Boundary region

Set partition

Integer partition

Concave-convex shape

Integral representation

AZ-identity

Optimization

ABSTRACT

Let n and m be integers, $n > m \geq 2$, and let $\mathcal{A} = \{A_1, \dots, A_m\}$ be a partition of $[n]$, where $[n] = \{1, \dots, n\}$. For a subset X of $[n]$, its $\mathcal{A}$ -boundary region $\mathcal{A}(X)$ is defined to be the union of those blocks A_i of $\mathcal{A}$ for which $A_i \cap X \neq \emptyset$ and $A_i \cap ([n] \setminus X) \neq \emptyset$. Partitions $\mathcal{A}$ of $[n]$ into m parts are determined such that, in a first case, the expected absolute cardinality and, in a second case, the expected relative cardinality of the $\mathcal{A}$ -boundary region of a randomly chosen subset of $[n]$ is minimal and maximal, respectively. The problem can be reduced to an optimization problem for integer partitions of n . In the most difficult case, the concave-convex shape of the corresponding weight function as well as several other inequalities are proved using an integral representation of the weight function. In a modified third setting, there is an analogon to the AZ-identity. The study is motivated by rough set theory.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Let n and m be integers, $n > m \geq 2$, and let $\mathcal{A} = \{A_1, \dots, A_m\}$ be a partition of $[n]$, where $[n] = \{1, \dots, n\}$. For a subset X of $[n]$, the $\mathcal{A}$ -lower approximation of X is defined by

$$\mathcal{A}^-(X) = \bigcup_{i: A_i \subseteq X} A_i$$

and the $\mathcal{A}$ -upper approximation of X is defined by

$$\mathcal{A}^+(X) = \bigcup_{i: A_i \cap X \neq \emptyset} A_i.$$

Obviously, $\mathcal{A}^-(X) \subseteq X \subseteq \mathcal{A}^+(X)$ for all X . The set

$$\mathcal{A}(X) = \mathcal{A}^+(X) \setminus \mathcal{A}^-(X)$$

is called the $\mathcal{A}$ -boundary region of X .

These notions stem from *rough set theory* that was initiated by Pawlak in [5] (see also [6]) and is built among other things on the following basic approach. Let P be a set of functions with domain $[n]$ and interpret the elements of $[n]$ as objects and the elements of P as attributes. Two objects i and j are said to be *equivalent* if they have the same attribute-values, i.e., if $p(i) = p(j)$ for all $p \in P$. Obviously, this defines an equivalence relation and hence there is a partition $\mathcal{A}$ of $[n]$ into

* Corresponding author.

E-mail addresses: konrad.engel@uni-rostock.de (K. Engel), tdt@hcmus.edu.vn (T.D. Thu).

equivalence classes. Now assume that there is given a set X of objects, but only the attribute-values can be used for the description of X . Then $\mathcal{A}^-(X)$ consists of those objects that can be positively identified as members of X and $\mathcal{A}^+(X)$ consists of those objects that might be members of X . Thus $|\mathcal{A}(X)|$ can be interpreted as a measure of uncertainty of X generated by the attribute-values. If $\mathcal{A}(X) = \emptyset$ then X is called a *crisp set*, otherwise it is called a *rough set*, but here we consider crisp sets also as degenerated rough sets.

In the following, we consider a uniformly distributed random variable $\mathbf{X}$ whose values are the members of the power set of $[n]$, i.e., $P(\mathbf{X} = X) = \frac{1}{2^n}$ for all $X \subseteq [n]$. If the partition $\mathcal{A}$ is fixed, then $|\mathcal{A}(\mathbf{X})|$ and $\frac{1}{|X|}|\mathcal{A}(\mathbf{X})|$ are two functions of the random variable $\mathbf{X}$ (where $\frac{1}{|X|}|\mathcal{A}(X)|$ is defined to be 0 if $X = \emptyset$).

The main aim of the paper is to determine those partitions $\mathcal{A}$ of $[n]$ into m parts that maximize or minimize the expected value of these two functions of $\mathbf{X}$, i.e.,

$$\mu_1(\mathcal{A}) = \frac{1}{2^n} \sum_{X \subseteq [n]} |\mathcal{A}(X)|,$$

which can be interpreted as an average *absolute uncertainty*, and

$$\mu_2(\mathcal{A}) = \frac{1}{2^n} \sum_{\emptyset \neq X \subseteq [n]} \frac{1}{|X|} |\mathcal{A}(X)|,$$

which can be interpreted as an average *relative uncertainty*.

In view of a nice identity, we study also a third case. Here, for fixed $k > 0$, $\mathbf{X}$ is a uniformly distributed random variable whose values are the k -element subsets of $[n]$, i.e., $P(\mathbf{X} = X) = \frac{1}{\binom{n}{k}}$ for all $X \in \binom{[n]}{k}$. Then the expected value of the function $\frac{1}{|X|}|\mathcal{A}(\mathbf{X})|$ is given by

$$\mu_{3,k}(\mathcal{A}) = \frac{1}{\binom{n}{k}} \sum_{X \in \binom{[n]}{k}} \frac{1}{k} |\mathcal{A}(X)|.$$

Now the aim is to optimize the average value of all these expected values, i.e., $\mu_3(\mathcal{A}) = \frac{1}{n} \sum_{k=1}^n \mu_{3,k}(\mathcal{A})$ which can be briefly written as

$$\mu_3(\mathcal{A}) = \frac{1}{n} \sum_{\emptyset \neq X \subseteq [n]} \frac{1}{\binom{n}{|X|}|X|} |\mathcal{A}(X)|.$$

This number can be interpreted as the average of the *average relative uncertainty for fixed cardinality of X* .

We note that allowing all partitions of $[n]$ instead of partitions of $[n]$ into m parts would make the problem trivial.

2. Reduction to an optimization problem for integer partitions

Since the factors $\frac{1}{2^n}$ (or $\frac{1}{n}$, respectively) do not have influence, we define

$$f_1(\mathcal{A}) = \sum_{X \subseteq [n]} |\mathcal{A}(X)|,$$

$$f_2(\mathcal{A}) = \sum_{\emptyset \neq X \subseteq [n]} \frac{1}{|X|} |\mathcal{A}(X)|,$$

$$f_3(\mathcal{A}) = \sum_{\emptyset \neq X \subseteq [n]} \frac{1}{\binom{n}{|X|}|X|} |\mathcal{A}(X)|$$

and further

$$f_j = \min\{f_j(\mathcal{A}) : \mathcal{A} \text{ is a partition of } [n] \text{ into } m \text{ parts}\},$$

$$\bar{f}_j = \max\{f_j(\mathcal{A}) : \mathcal{A} \text{ is a partition of } [n] \text{ into } m \text{ parts}\}.$$

Partitions $\mathcal{A}$ of $[n]$ with $f_j(\mathcal{A}) = f_j$ or $f_j(\mathcal{A}) = \bar{f}_j$ are called *j-minimal* and *j-maximal*, respectively.

With a partition $\mathcal{A} = \{A_1, \dots, A_m\}$ we associate its block sizes $a_i = |A_i|$, $i = 1, \dots, m$. It is not difficult to see that by symmetry the values $f_j(\mathcal{A})$ only depend on the block sizes of $\mathcal{A}$ and not on the concrete partition. Indeed, let $w : \{0, \dots, n\} \rightarrow \mathbb{R}$ be any (weight) function, let

$$f_w(\mathcal{A}) = \sum_{X \subseteq [n]} w(|X|) |\mathcal{A}(X)|$$

and let $F_w : \{0, \dots, n\} \rightarrow \mathbb{R}$ be defined (with $\binom{n}{k} = 0$ if $k < 0$) by

$$F_w(a) = \sum_{k=0}^n \left(\binom{n-a}{k} + \binom{n-a}{k-a} \right) w(k).$$

Now the following lemma converts the boundary optimization problem for rough sets into an optimization problem for integer partitions.

Lemma 1. *We have*

$$f_w(\mathcal{A}) = n \sum_{k=0}^n \binom{n}{k} w(k) - \sum_{i=1}^m a_i F_w(a_i).$$

Proof. We have

$$\begin{aligned} f_w(\mathcal{A}) &= \sum_{X \subseteq [n]} w(|X|) (|\mathcal{A}^+(X)| - |\mathcal{A}^-(X)|) \\ &= \sum_{X \subseteq [n]} w(|X|) \left(\sum_{i: A_i \cap X \neq \emptyset} |A_i| - \sum_{i: A_i \subseteq X} |A_i| \right) \\ &= \sum_{X \subseteq [n]} w(|X|) \left(n - \sum_{i: A_i \cap X = \emptyset} |A_i| - \sum_{i: A_i \subseteq X} |A_i| \right) \\ &= n \sum_{k=0}^n \binom{n}{k} w(k) - \sum_{i=1}^m a_i \left(\sum_{X: X \subseteq [n] \setminus A_i} w(|X|) + \sum_{X: A_i \subseteq X} w(|X|) \right) \\ &= n \sum_{k=0}^n \binom{n}{k} w(k) - \sum_{i=1}^m a_i \left(\sum_{k=0}^n \left(\binom{n-a_i}{k} + \binom{n-a_i}{k-a_i} \right) w(k) \right) \\ &= n \sum_{k=0}^n \binom{n}{k} w(k) - \sum_{i=1}^m a_i F_w(a_i). \quad \square \end{aligned}$$

Let $\mathcal{P}(n, m)$ be the set of all integer partitions $\mathbf{a} = (a_1, a_2, \dots, a_m)$ of the integer n into m parts, i.e., $a_1, a_2, \dots, a_m$ are positive integers with $a_1 + \dots + a_m = n$. With the concrete weight functions $w_j : \{0, \dots, n\} \rightarrow \mathbb{R}$, where

$$\begin{aligned} w_1(k) &= 1 \quad \text{for all } k, \\ w_2(k) &= \begin{cases} 0, & \text{if } k = 0, \\ \frac{1}{k}, & \text{otherwise,} \end{cases} \\ w_3(k) &= \begin{cases} 0, & \text{if } k = 0, \\ \frac{1}{\binom{n}{k}k}, & \text{otherwise,} \end{cases} \end{aligned}$$

we obtain from Lemma 1

$$\begin{aligned} \underline{f}_j &= n \sum_{k=0}^n \binom{n}{k} w_j(k) - \max \left\{ \sum_{i=1}^m a_i F_{w_j}(a_i) : \mathbf{a} \in \mathcal{P}(n, m) \right\}, \\ \bar{f}_j &= n \sum_{k=0}^n \binom{n}{k} w_j(k) - \min \left\{ \sum_{i=1}^m a_i F_{w_j}(a_i) : \mathbf{a} \in \mathcal{P}(n, m) \right\}. \end{aligned}$$

We call an integer partition $\mathbf{a} \in \mathcal{P}(n, m)$ *j-minimal* and *j-maximal* if $\sum_{i=1}^m a_i F_{w_j}(a_i)$ attains the minimum and maximum, respectively.

Corollary 1. *The set partition $\mathcal{A}$ is j-minimal (resp. j-maximal) iff the block sizes of $\mathcal{A}$ form a j-maximal (resp. j-minimal) integer partition $\mathbf{a} \in \mathcal{P}(n, m)$.*

Thus we restrict ourselves to the determination of *j-optimal* integer partitions.

We mention that, for arbitrary weight functions F (instead of the special weight functions F_{w_j}) and arbitrary number of items (instead of fixed number of items), a dynamic programming algorithm for the determination of optimal integer

partitions is given in [4]. Moreover, [4] contains bounds for the number of pairwise different items in an optimal integer partition. But for our special cases of functions F_{w_j} we may provide explicit solutions. For more information on integer partitions see [2].

In the following we present integer partitions from $\mathcal{P}(n, m)$ also in the form $1^{\lambda_1} 2^{\lambda_2} \dots n^{\lambda_n}$ which means that there are exactly λ_i items $i, i = 1, \dots, n$, and, equivalently,

$$\sum_{i=1}^n i \lambda_i = n \text{ and } \sum_{i=1}^n \lambda_i = m \text{ with non-negative integers } \lambda_1, \dots, \lambda_n.$$

Moreover, items of the form i^0 will be mostly omitted.

3. Main results

The main results are given by the following three theorems.

Theorem 1.

(a) An integer partition of $\mathcal{P}(n, m)$ is 1-minimal if it is of the form

$$\begin{cases} 1^{m-1}(n-m+1)^1 & \text{if } n-m \leq 5, \\ q^{m-r}(q+1)^r & \text{if } n-m > 5, n > 5m, n = qm + r, 0 \leq r < m, \\ 1^{m-\frac{n-m}{4}} 5^{\frac{n-m}{4}} & \text{if } n-m > 5, n \leq 5m \text{ and } m-n \equiv 0 \pmod{4}, \\ 1^{m-\frac{n-m+1}{4}} 4^1 5^{\frac{n-m-3}{4}} & \text{if } n-m > 5, n \leq 5m \text{ and } m-n \equiv 1 \pmod{4}, \\ 1^{m-\frac{n-m+2}{4}} 4^2 5^{\frac{n-m-6}{4}} & \text{if } n-m > 5, n \leq 5m \text{ and } m-n \equiv 2 \pmod{4}, \\ 1^{m-\frac{n-m-1}{4}} 5^{\frac{n-m-5}{4}} 6^1 & \text{if } n-m > 5, n \leq 5m \text{ and } m-n \equiv 3 \pmod{4}. \end{cases}$$

(b) An integer partition of $\mathcal{P}(n, m)$ is 1-maximal if it is of the form

$$\begin{cases} 2^{m-1}(n-2m+2)^1 & \text{if } n \geq 2m, \\ 1^{2m-n} 2^{n-m} & \text{if } n < 2m. \end{cases}$$

Theorem 2. Let $n \geq 11$.

(a) An integer partition of $\mathcal{P}(n, m)$ is 2-minimal if it is of the form given in part (a) of Theorem 1.

(b) An integer partition of $\mathcal{P}(n, m)$ is 2-maximal if it is of the form

$$\begin{cases} 2^{m-1}(n-2m+2)^1 & \text{if } n > 2m+2, \\ 2^{m-2} 3^2 & \text{if } n = 2m+2, \\ 2^{m-1} 3^1 & \text{if } n = 2m+1, \\ 1^{2m-n} 2^{n-m} & \text{if } n \leq 2m. \end{cases}$$

Theorem 3.

(a) An integer partition of $\mathcal{P}(n, m)$ is 3-minimal if it is of the form

$$1^{m-1}(n-m+1)^1.$$

(b) An integer partition of $\mathcal{P}(n, m)$ is 3-maximal if it is of the form

$$q^{m-r}(q+1)^r, \text{ where } n = qm + r, 0 \leq r < m.$$

4. Optimization for integer partitions

It will turn out that our functions F_{w_j} have a concave-convex shape. Therefore we present an elementary optimality criterion that is at least for the special case $c = 1$ folklore. Let $g : [n] \rightarrow \mathbb{R}$ be any function. The integer partition $\mathbf{a} \in \mathcal{P}(n, m)$ is called g -minimal (resp. g -maximal) if the objective function $\sum_{i=1}^m g(a_i)$ attains the minimum (resp. maximum) at $\mathbf{a}$.

Theorem 4. Assume that there is a number $c \in \{2, \dots, n\}$ such that

$$2g(a) > g(a-1) + g(a+1) \text{ for all } a \in \{2, \dots, c-1\}, \quad (1)$$

$$2g(a) \leq g(a-1) + g(a+1) \text{ if } a = c, \quad (2)$$

$$2g(a) < g(a-1) + g(a+1) \text{ for all } a \in \{c+1, \dots, n-1\}. \quad (3)$$

- (a) If $\mathbf{a}^* \in \mathcal{P}(n, m)$ is a g -minimal partition then it has the following properties: $\mathbf{a}^*$ does not contain two items from $\{c - 1, \dots, n\}$ with absolute difference ≥ 2 , with the exception that items $c - 1$ and $c + 1$ may exist if (2) is satisfied with equality. $\mathbf{a}^*$ does not contain two items from $\{2, \dots, c - 1\}$.
- (b) If $\mathbf{a}^{**} \in \mathcal{P}(n, m)$ is a g -maximal partition then it has the following properties: $\mathbf{a}^{**}$ does not contain two items from $\{1, \dots, c\}$ with absolute difference ≥ 2 . $\mathbf{a}^{**}$ does not contain two items from $\{c, \dots, n - 1\}$, with the exception that two items c may exist if (2) is satisfied with equality.

Proof. We prove only part (a) because part (b) can be proved analogously. Let briefly $g(\mathbf{a}) = \sum_{i=1}^m g(a_i)$ and let $\mathbf{a}^*$ be a g -minimal partition. Assume that $\mathbf{a}^*$ contains two items a_i, a_j from $\{c - 1, \dots, n\}$ with $a_j - a_i \geq 2$ (where $a_i = c - 1$ and $a_j = c + 1$ are not considered if (2) is satisfied with equality). Then let $\mathbf{a}'$ be the partition that can be obtained from $\mathbf{a}^*$ by incrementing a_i and decrementing a_j . Then

$$g(\mathbf{a}') - g(\mathbf{a}^*) = (g(a_j - 1) - g(a_j)) - (g(a_i) - g(a_i + 1)) < 0$$

since by (3) for all $a \in \{c + 1, \dots, n\}$

$$g(a - 1) - g(a) > g(a) - g(a + 1) > g(a + 1) - g(a + 2) > \dots$$

Hence $\mathbf{a}^*$ cannot be g -minimal, a contradiction.

Assume that $\mathbf{a}^*$ contains two items a_i, a_j from $\{2, \dots, c - 1\}$ with $a_j \geq a_i$. Let $\mathbf{a}'$ be the partition that can be obtained from $\mathbf{a}^*$ by decrementing a_i and incrementing a_j . Then

$$g(\mathbf{a}') - g(\mathbf{a}^*) = (g(a_j + 1) - g(a_j)) - (g(a_i) - g(a_i - 1)) < 0$$

since by (1) for all $a \in \{2, \dots, c - 1\}$

$$g(a) - g(a - 1) > g(a + 1) - g(a) > g(a + 2) - g(a + 1) > \dots > g(c) - g(c - 1).$$

Hence $\mathbf{a}^*$ cannot be g -minimal, a contradiction. $\square$

For a k -tuple $(a_{i_1}, \dots, a_{i_k})$ of items from $\mathbf{a}$ let briefly $g(a_{i_1}, \dots, a_{i_k}) = \sum_{l=1}^k g(a_{i_l})$. In order to obtain contradictions we replace such a k -tuple by some k -tuple $(a'_{i_1}, \dots, a'_{i_k})$ of items such that $a_{i_1} + \dots + a_{i_k} = a'_{i_1} + \dots + a'_{i_k}$ and $g(a_{i_1}, \dots, a_{i_k}) <$ (resp. $>$) $g(a'_{i_1}, \dots, a'_{i_k})$.

5. Proof of Theorem 1

We have

$$F_{w_1}(a) = \sum_{k=0}^n \left(\binom{n-a}{k} + \binom{n-a}{k-a} \right) = 2^{n-a+1}.$$

Clearly we may ignore the factor 2^{n+1} and thus we work with the function

$$g(a) = \frac{a}{2^a} \quad \left(= \frac{1}{2^{n+1}} a F_{w_1}(a) \right).$$

It is easy to check that (1), (2) (with equality) and (3) of Theorem 4 are satisfied with $c = 3$. Moreover, note that

$$g(1) = g(2) > g(3) > g(4) > \dots \quad (4)$$

Part (a): Let $\mathbf{a}^*$ be a 1-minimal partition. In addition, we choose $\mathbf{a}^*$ in such a way that it has a minimum number of items 4 with respect to all 1-minimal partitions. Then, by Theorem 4(a), $\mathbf{a}^*$ does not contain two items 2. Moreover, by the special choice of $\mathbf{a}^*$ it does not contain a pair (2, 4) because it could be replaced by (3, 3) which contradicts the 1-minimality and the minimality of the number of items 4. Thus, by Theorem 4(a), $\mathbf{a}^*$ has the form $1^i b^j (b + 1)^k$ with some integer $b \geq 2$, i.e.,

$$i + j + k = m \text{ and } i + bj + (b + 1)k = n. \quad (5)$$

We may assume that $k \geq 1$ because a partition $1^i b^j (b + 1)^0$ can also be considered as a partition $1^i (b - 1)^0 ((b - 1) + 1)^j$.

Since the reasoning in the proof of Theorem 2 will be analogous we point out that we use the following inequalities (recall that e.g. $g(2, 3) = g(2) + g(3)$):

$$g(2, 3) > g(1, 4), \quad (6)$$

$$g(3, 3) > g(1, 5), \quad (7)$$

$$g(3, 4) > g(1, 6), \quad (8)$$

$$g(1, 2k - 1) > g(k, k) \text{ for all } k \geq 4, \quad (9)$$

$$g(1, 2k) > g(k, k + 1) \text{ for all } k \geq 4, \quad (10)$$

$$g(1, 6, 6) > g(4, 4, 5), \quad (11)$$

$$g(4, 4, 4) \geq g(1, 5, 6) \text{ (in fact } g(4, 4, 4) = g(1, 5, 6)). \quad (12)$$

These inequalities can be verified for $g(a) = \frac{a}{2^a}$ by elementary computations. Concerning (9) (and analogously (10)) note that $g(1, 2k-1) > g(1) = \frac{1}{2} = 2g(4) \geq g(k, k)$.

Case 1. $n - m \leq 5$. We have to show that $j + k \leq 1$. Assume that $j + k \geq 2$. By (5),

$$5 \geq n - m = (b-1)j + bk \geq (b-1)(j+k) \geq (b-1)2$$

and hence $b \leq 3$ and, moreover, $k = 1$ if $b = 3$. Thus $\mathbf{a}^*$ contains a pair of items of the form (2, 3) or (3, 3) or (3, 4). But such a pair could be replaced by (1, 4), (1, 5) or (1, 6), respectively, which contradicts the 1-minimality of $\mathbf{a}^*$ by (6), (7) and (8), respectively.

Case 2. $n - m > 5$ and $n > 5m$. We have to show that $i = 0$.

Assume that $i > 0$, i.e., $\mathbf{a}^*$ contains an item 1. The partition $\mathbf{a}^*$ cannot contain an item $d \geq 7$ because the pair (1, d) could be replaced by $(\frac{d+1}{2}, \frac{d+1}{2})$ if d is odd and by $(\frac{d}{2}, \frac{d}{2} + 1)$ if d is even which contradicts the 1-minimality of $\mathbf{a}^*$ by (9) and (10). Thus $b \leq 5$.

Assume that $\mathbf{a}^*$ contains at most one item 6. Then by (5)

$$5m - n = 4i + (5-b)j + (5-(b+1))k \geq 4 + 0 - 1 > 0,$$

a contradiction to $n > 5m$. Thus $\mathbf{a}^*$ contains an item 1 and two items 6. But (1, 6, 6) can be replaced by (4, 4, 5) which contradicts the 1-minimality of $\mathbf{a}^*$ by (11).

Case 3. $n - m > 5$ and $n \leq 5m$. If $n - m = 6$, then, similarly to Case 1, $\mathbf{a}^*$ contains a pair of items (2, 3) or (3, 3) or (3, 4) or (4, 4). Since the first three pairs yield a contradiction $\mathbf{a}^*$ has the form $1^{m-2}4^2$ as asserted.

Thus let $n - m > 6$.

First note that $\mathbf{a}^*$ cannot contain three items 4 because the triple (4, 4, 4) could be replaced by (1, 5, 6) which contradicts the 1-minimality and the minimality of the number of items 4 of $\mathbf{a}^*$ by (12). Now we show that

$$b = 4 \text{ or } (b = 5 \text{ and } k = 1). \quad (13)$$

In order to show that $b \geq 4$ assume that $b \leq 3$. Then $j + k > 2$ because otherwise $6 < n - m = (b-1)j + bk \leq b(j+k) \leq 6$. Thus $\mathbf{a}^*$ contains a pair of items of the form (2, 3) or (3, 3) or (3, 4) which leads as in Case 1 to a contradiction.

Assume that $b \geq 6$. Then $\mathbf{a}^*$ must contain an item 1 in view of $n \leq 5m$ and an item $d = (b+1) \geq 7$. This leads as in Case 2 to a contradiction.

Now let $b = 5$. If $i = 0$ then in view of $n \leq 5m$ necessarily $n = 5m$ and hence $\mathbf{a}^*$ has the form $1^0 4^0 5^m$, i.e., $b = 4$, a contradiction. Thus $i > 0$, i.e., $\mathbf{a}^*$ contains an item 1. Then $\mathbf{a}^*$ cannot contain two items 6 because otherwise (1, 6, 6) could be replaced by (4, 4, 5) which contradicts the 1-minimality of $\mathbf{a}^*$ by (11). Hence $k = 1$ and (13) is proved.

Thus $\mathbf{a}^*$ has necessarily one of the following forms: $1^i 5^k$, $1^i 4^1 5^k$, $1^i 4^2 5^k$, $1^i 5^j 6^1$.

From (5) it follows that $i = m - \frac{n-m+j}{b}$ and $k = \frac{n-m-(b-1)j}{b}$. Consequently:

The form $1^i 5^k$ requires $m - n \equiv 0 \pmod{4}$ and is more precisely given by $1^{m-\frac{n-m}{4}} 5^{\frac{n-m}{4}}$.

The form $1^i 4^1 5^k$ requires $m - n \equiv 1 \pmod{4}$ and is more precisely given by $1^{m-\frac{n-m+1}{4}} 4^1 5^{\frac{n-m-3}{4}}$.

The form $1^i 4^2 5^k$ requires $m - n \equiv 2 \pmod{4}$ and is more precisely given by $1^{m-\frac{n-m+2}{4}} 4^2 5^{\frac{n-m-6}{4}}$.

The form $1^i 5^j 6^1$ has $1 = k = \frac{n-m-(b-1)j}{b}$ which leads to the precise form $1^{m-\frac{n-m-1}{4}} 5^{\frac{n-m-5}{4}} 6^1$ and requires $m - n \equiv 3 \pmod{4}$.

Thus part (a) is proved.

Part (b): Let $\mathbf{a}^{**}$ be a 1-maximal partition. In addition, we choose $\mathbf{a}^{**}$ in such a way that it has a minimum number of items 3 with respect to all 1-maximal partitions. Then, by Theorem 4(b), $\mathbf{a}^{**}$ has the form $1^i 2^j b^k$ with some integer $b \geq 3$ and $k \leq 1$.

Case 1. $n > 2m$. Then $k = 1$. We have to show that $i = 0$. Assume that $i \geq 1$. By (4), $g(1, b) < g(2, b-1)$, which contradicts the 1-maximality of $\mathbf{a}^{**}$.

Case 2. $n \leq 2m$. We have to show that $k = 0$. Assume that $k \geq 1$. If $i = 0$ then $n = 2j + bk > 2(j+k) = 2m$, a contradiction. Thus $i \geq 1$. As in Case 1, (1, b) can be replaced by (2, $b-1$), a contradiction. $\square$

6. Proof of Theorem 2

We have

$$F_{w_2}(a) = \sum_{k=1}^n \left(\binom{n-a}{k} + \binom{n-a}{k-a} \right) \frac{1}{k}.$$

The main idea is to use an integral representation of $F_{w_2}(a)$ and to prove inequalities “under the integral”.

Let

$$p_a^+(t) = \frac{1}{t} \left((1+t)^{n-a} - 1 \right),$$

$$p_a^-(t) = t^{a-1} (1+t)^{n-a}.$$

The binomial theorem and a simple evaluation of the integrals yield the following representation.

Lemma 2. *We have*

$$F_{w_2}(a) = \int_0^1 (p_a^+(t) + p_a^-(t)) dt.$$

Consequently, for the proof of [Theorem 2](#), we work with the function

$$g(a) = \int_0^1 a (p_a^+(t) + p_a^-(t)) dt.$$

In order to prove inequalities, we need the following estimation.

Lemma 3. *Let $\varphi(t)$ and $\varrho(t)$ be continuous real functions on $[0, 1]$. Let $\varphi(t)$ be positive and increasing and suppose that there is some root $x_0 \in (0, 1)$ of $\varrho(t)$ such that $\varrho(t) < 0$ for all $t \in [0, x_0]$ and $\varrho(t) > 0$ for all $t \in (x_0, 1]$. If*

$$\int_0^1 \varrho(t) dt > 0 \quad (\text{resp. } \int_0^1 \varrho(t) dt \geq 0) \quad (14)$$

then

$$\int_0^1 \varphi(t) \varrho(t) dt > 0 \quad (\text{resp. } \int_0^1 \varphi(t) \varrho(t) dt \geq 0). \quad (15)$$

Proof. We have

$$\begin{aligned} \int_0^1 \varphi(t) \varrho(t) dt &= \int_0^{x_0} \varphi(t) \varrho(t) dt + \int_{x_0}^1 \varphi(t) \varrho(t) dt \\ &\geq \int_0^{x_0} \varphi(x_0) \varrho(t) dt + \int_{x_0}^1 \varphi(x_0) \varrho(t) dt \\ &= \varphi(x_0) \int_0^1 \varrho(t) dt \\ &> 0 \quad (\text{resp. } \geq 0). \quad \square \end{aligned}$$

First we study the monotonicity of g .

Lemma 4. *Let $n \geq 6$. We have*

$$\begin{aligned} g(a+1) &> g(a) \text{ if } a = 1, \\ g(a+1) &< g(a) \text{ if } a \in \{2, \dots, n-1\}. \end{aligned}$$

Proof. In fact, the functions p_a^+ , p_a^- , g depend also on n , hence we write here more detailed $p_{a,n}^+$, $p_{a,n}^-$, g_n . Moreover, let

$$\begin{aligned} p_{a,n}(t) &= a (p_{a,n}^+(t) + p_{a,n}^-(t)), \\ q_{a,n}(t) &= p_{a,n+1}(t) - p_{a,n}(t), \\ d_{a,n}(t) &= p_{a+1,n}(t) - p_{a,n}(t). \end{aligned}$$

Note that

$$\begin{aligned} g_n(a) &= \int_0^1 p_{a,n}(t) dt, \\ q_{a,n} &= a(1+t)^{n-a}(1+t^a) \end{aligned}$$

and

$$g_n(a+1) - g_n(a) = \int_0^1 d_{a,n}(t) dt. \quad (16)$$

By definition,

$$\begin{aligned} p_{a,a+1} &= a + at^{a-1}(1+t), \\ p_{a+1,a+1} &= (a+1)t^a, \end{aligned}$$

and hence, for all $t \in [0, 1]$,

$$d_{a,a+1} = -a + t^{a-1}(t-a) < 0,$$

which implies together with (16) that

$$g_{a+1}(a+1) - g_{a+1}(a) < 0. \quad (17)$$

For $n \geq a+1$ we have

$$\begin{aligned} d_{a,n+1}(t) - d_{a,n}(t) &= (p_{a+1,n+1}(t) - p_{a,n+1}(t)) - (p_{a+1,n}(t) - p_{a,n}(t)) \\ &= q_{a+1,n}(t) - q_{a,n}(t) \\ &= (a+1)(1+t)^{n-a-1}(1+t^{a+1}) - a(1+t)^{n-a}(1+t^a) \\ &= (1+t)^{n-a-1}((1-at) + t^a(t-a)). \end{aligned}$$

In the case $a = 1$, we have for all $t \in (0, 1)$

$$d_{1,n+1}(t) - d_{1,n}(t) = (1+t)^{n-2}(1-t)^2 > 0$$

and by (16)

$$(g_{n+1}(2) - g_{n+1}(1)) - (g_n(2) - g_n(1)) > 0.$$

Since $g_6(2) - g_6(1) = \frac{29}{60} > 0$ it follows that $g_n(2) > g_n(1)$ if $n \geq 6$.

In the case $a \geq 2$, we have

$$d_{a,n+1}(t) - d_{a,n}(t) \geq (1+t)^{n-a-1}((1-2t) + t^a(t-a)).$$

By Lemma 3 applied to $\varphi(t) = (1+t)^{n-a-1}$ and $\varrho(t) = 2t-1$ and in view of $t^a(t-a) < 0$ for all $t \in (0, 1]$ we obtain easily that

$$\int_0^1 (1+t)^{n-a-1}((1-2t) + t^a(t-a)) dt < 0$$

and hence

$$(g_{n+1}(a+1) - g_{n+1}(a)) - (g_n(a+1) - g_n(a)) < 0,$$

which together with (17) implies that $g_n(a+1) < g_n(a)$ if $a \geq 2$. $\square$

Let $\mathbf{a} = (a_1, \dots, a_k)$ and $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_k)$ be k -tuples of integers. To prove further inequalities, we have to evaluate sums of the form $\sum_{i=1}^k \alpha_i g(a_i)$. For this purpose, we introduce the polynomials

$$\begin{aligned} q_{\boldsymbol{\alpha}, \mathbf{a}}^+(t) &= \sum_{i=1}^k \alpha_i a_i (1+t)^{a_k-a_i}, \\ q_{\boldsymbol{\alpha}, \mathbf{a}}^-(t) &= \sum_{i=1}^k \alpha_i a_i t^{a_i-a_1} (1+t)^{a_k-a_i}, \end{aligned}$$

but, for a simpler representation, we omit the indices $\boldsymbol{\alpha}, \mathbf{a}$ and write simply $q^+(t)$ and $q^-(t)$.

Note that

$$q^+(t) = t^{a_k-a_1} q^-(1/t). \quad (18)$$

Lemma 5. Let $\sum_{i=1}^k \alpha_i a_i = 0$. Then

$$\sum_{i=1}^k \alpha_i g(a_i) = \int_0^1 (1+t)^{n-a_k} (t^{a_k-a_1-1} q^-(1/t) + t^{a_1-1} q^-(t)) dt.$$

Proof. We have

$$\begin{aligned} \sum_{i=1}^k \alpha_i a_i (p_{a_i}^+(t) + p_{a_i}^-(t)) &= \sum_{i=1}^k \alpha_i a_i \frac{1}{t} ((1+t)^{n-a_i} - 1) + \alpha_i a_i t^{a_i-1} (1+t)^{n-a_i} \\ &= (1+t)^{n-a_k} \sum_{i=1}^k \alpha_i a_i \frac{1}{t} (1+t)^{a_k-a_i} + \alpha_i a_i t^{a_i-1} (1+t)^{a_k-a_i} \\ &= (1+t)^{n-a_k} \left(\frac{1}{t} q^+(t) + t^{a_1-1} q^-(t) \right) \\ &= (1+t)^{n-a_k} (t^{a_k-a_1-1} q^-(1/t) + t^{a_1-1} q^-(t)). \quad \square \end{aligned}$$

In the following let

$$r(t) = t^{a_k - a_1 - 1} q^-(1/t) + t^{a_1 - 1} q^-(t),$$

which implies that

$$\sum_{i=1}^k \alpha_i g(a_i) = \int_0^1 (1+t)^{n-a_k} r(t) dt. \quad (19)$$

Lemma 6.

- (a) If $r(t) \geq 0$ for all $t \in [0, 1]$ and $r(t)$ is not the zero polynomial then $\sum_{i=1}^k \alpha_i g(a_i) > 0$ for all $n \geq a_k$.
 (b) If there is some root $x_0 \in (0, 1)$ of $r(t)$ such that $r(t) < 0$ for all $t \in [0, x_0)$ and $r(t) > 0$ for all $t \in (x_0, 1]$ and if there is some integer l such that $\int_0^1 (1+t)^l r(t) dt > 0$ then $\sum_{i=1}^k \alpha_i g(a_i) > 0$ for all $n \geq a_k + l$.

Proof. Part (a) is a trivial consequence of (19) and part (b) follows from Lemma 3 applied to $\varphi(t) = (1+t)^{n-a_k-l}$ and $\varrho(t) = (1+t)^l r(t)$. $\square$

Lemma 7. Let $n \geq 6$. We have

$$2g(a) > g(a-1) + g(a+1) \text{ if } a \in \{2, 3\},$$

$$2g(a) < g(a-1) + g(a+1) \text{ if } a \in \{4, \dots, n-1\}.$$

Proof. We have to estimate $g(a-1) - 2g(a) + g(a+1)$ and hence we work with $\mathbf{a} = (a-1, a, a+1)$ and $\boldsymbol{\alpha} = (1, -2, 1)$ and apply Lemma 6. We have

$$q^-(t) = (a-1)(1+t)^2 - 2at(1+t) + (a+1)t^2 = a - 2t - 1,$$

$$r(t) = t^{a-2}(a-2t-1) + (a-1)t - 2.$$

If $a = 2$ then $r(t) = -(t+1) < 0$ for all $t \in [0, 1]$, if $a = 3$ then $r(t) = -2(t-1)^2 < 0$ for all $t \in [0, 1)$ and hence by Lemma 6(a) (applied to $-r(t)$)

$$g(a-1) - 2g(a) + g(a+1) < 0 \quad \text{if } a \in \{2, 3\}.$$

If $a = 4$ then $r(t) = -2t^3 + 3t^2 + 3t - 2$, $r(t) < 0$ for all $t \in [0, \frac{1}{2})$, $r(t) > 0$ for all $t \in (\frac{1}{2}, 1]$, and $\int_0^1 (1+t)^1 r(t) dt = \frac{7}{20} > 0$. By Lemma 6(b) (note that $n - (a+1) \geq 6 - 5 = 1$)

$$g(a-1) - 2g(a) + g(a+1) > 0 \quad \text{if } a = 4.$$

Now let $a \geq 5$. Clearly $a - 2t - 1 > 0$ for all $t \in [0, 1]$ and hence

$$\int_0^1 r(t) dt > \int_0^1 (a-1)t - 2 \geq 0.$$

By Lemma 6(a)

$$g(a-1) - 2g(a) + g(a+1) > 0 \quad \text{if } a \in \{5, \dots, n-1\}. \quad \square$$

The next lemma summarizes several further inequalities that are needed in the following.

Lemma 8. Let $n \geq 13$. Then

$$g(1, 7) > g(4, 4), \quad (20)$$

$$g(1, 8) > g(4, 5), \quad (21)$$

$$g(2, 3) > g(1, 4), \quad (22)$$

$$g(2, 5) > g(3, 4), \quad (23)$$

$$g(3, 3) > g(1, 5), \quad (24)$$

$$g(3, 4) > g(1, 6), \quad (25)$$

$$g(1, 6, 6) > g(4, 4, 5), \quad (26)$$

$$g(2, 2, 5) > g(3, 3, 3), \quad (27)$$

$$g(4, 4, 4) > g(1, 5, 6). \quad (28)$$

Proof. For the proof, we again apply Lemma 6, analogously to the proof of Lemma 7. For the sake of brevity we present the corresponding vectors $\mathbf{a}$ and $\boldsymbol{\alpha}$ in the form of a table.

Inequality	α	α
$g(1, 7) > g(4, 4)$	(1, 4, 7)	(1, -2, 1)
$g(1, 8) > g(4, 5)$	(1, 4, 5, 8)	(1, -1, -1, 1)
$g(2, 3) > g(1, 4)$	(1, 2, 3, 4)	(-1, 1, 1, -1)
$g(2, 5) > g(3, 4)$	(2, 3, 4, 5)	(1, -1, -1, 1)
$g(3, 3) > g(1, 5)$	(1, 3, 5)	(-1, 2, 1)
$g(3, 4) > g(1, 6)$	(1, 3, 4, 6)	(-1, 1, 1, -1)
$g(1, 6, 6) > g(4, 4, 5)$	(1, 4, 5, 6)	(1, -2, -1, 2)
$g(2, 2, 5) > g(3, 3, 3)$	(2, 3, 5)	(2, -3, 1)
$g(4, 4, 4) > g(1, 5, 6)$	(1, 4, 5, 6)	(-1, 3, -1, -1)

The corresponding terms $r(t)$ (if [Lemma 6\(a\)](#) can be applied) and additionally the values l , $\int_0^1 (1+t)^l r(t) dt$ and the lower bound $a_k + l$ for n (if [Lemma 6\(b\)](#) can be applied) can be computed and checked using a standard computer algebra system. They are given in the next table in the corresponding order.

$r(t)$	l	$\int_0^1 (1+t)^l r(t) dt$	$a_k + l$
$-17t^5 - 3t^4 + 27t^3 + 27t^2 - 3t - 17$	4	1639/360	11
$-23t^6 - 11t^5 + 35t^4 + 62t^3 + 35t^2 - 11t - 23$	0	23/42	8
$3t^2 - 2t + 1/3 \geq 0$	3	37/70	8
$-4t^3 + 5t^2 + 5t - 4$			
$7t^3 - 4t^2 - 4t + 7 \geq 0$			
$11t^4 - 2t^3 - 14t^2 - 2t + 11 \geq 0$			
$-15t^4 + 7t^3 + 20t^2 + 7t - 15$	7	247/660	13
$-6t^3 + 7t^2 + 7t - 6$	5	443/168	10
$13t^4 - 3t^3 - 20t^2 - 3t + 13 \geq 0$			

□

Lemma 9. Let $n \geq 11$. We have for all $k \geq 4$ with $2k \leq n$ resp. $2k < n$

$$g(1, 2k-1) > g(k, k), \quad (29)$$

$$g(1, 2k) > g(k, k+1). \quad (30)$$

Proof. Both inequalities can be proved by induction on k using (20) and (21) as the induction basis. As an example we show the step from k to $k+1$ for (29). We have for $k \geq 4$ by [Lemma 7](#)

$$\begin{aligned} g(2k+1) - g(2k-1) &= (g(2k+1) - g(2k)) + (g(2k) - g(2k-1)) \\ &\geq (g(k+1) - g(k)) + (g(k+1) - g(k)) \end{aligned}$$

and hence, by the induction hypothesis,

$$g(1, 2k+1) - g(k+1, k+1) \geq g(1, 2k-1) - g(k, k) > 0. \quad \square$$

Lemma 10. Let $n \geq 8$. Then for all a with $4 \leq a \leq n-3$

$$g(2) + g(a+1) > g(3) + g(a). \quad (31)$$

Proof. This inequality can be proved by induction on a using (23) as the induction basis. For the induction step from a to $a+1$ note that by [Lemma 7](#), $g(a+2) - g(a+1) > g(a+1) - g(a)$. Using the induction hypothesis we obtain

$$g(a+2) - g(a+1) + g(2) - g(3) > g(a+1) - g(a) + g(2) - g(3) > 0. \quad \square$$

Proof of Theorem 2. Part (a): The case $n = m+1$ is trivial, thus let $n > m+1$. Recall that $n \geq 11$. Let α^* be a 2-minimal partition. Assume that it contains an item 2. In view of $n > m+1$ it contains also an item $a \geq 2$. But the pair $(2, a)$ can be replaced by $(1, a+1)$, a contradiction to the 2-minimality of α^* by [Lemma 4](#). The conditions (1), (2) (with strict inequality) and (3) of [Theorem 4](#) are satisfied for $c = 4$ by [Lemma 7](#). Thus by [Theorem 4](#), α^* has the form $1^i b^j (b+1)^k$ with some $b \geq 3$.

Now we are in the same situation as in the proof of part (a) of [Theorem 1](#) (with the stronger inequality $b \geq 3$ instead of $b \geq 2$). The inequalities (6)–(12) are satisfied for the actual function $g(a) = aF_{w_2}(a)$ by [Lemmas 8](#) and [9](#). In fact, [Lemma 8](#) is stated for $n \geq 13$ instead of $n \geq 11$. But in the proof of [Lemma 8](#) we need the bound 13 only for the inequality (26) which is used to replace three items 1, 6, 6 by three items 4, 4, 5. But such a replacement is not required for $n = 11$ and $n = 12$ because $1 + 6 + 6 = 4 + 4 + 5 = 13$.

Thus the Cases 1–3 can be discussed exactly in the same way as in the proof of part (a) of [Theorem 1](#). Note that we do not need the additional assumption that $\mathbf{a}^*$ contains a minimal number of items 4 because the inequality (12) is here a strict inequality, see (28).

Part (b): Let $\mathbf{a}^{**}$ be a 2-maximal partition. It cannot contain a pair $(3, a)$ with $a \geq 4$ since it could be replaced by $(2, a+1)$ which contradicts the 2-maximality of $\mathbf{a}^{**}$ by [Lemma 10](#). Moreover, $\mathbf{a}^{**}$ cannot contain a pair $(1, a)$ with $a \geq 3$ because it could be replaced by $(2, a-1)$ ([Lemma 4](#)).

As for part (a), we may apply [Theorem 4](#) (but here part (b)) with $c = 4$ and strict inequality in (2). Thus $\mathbf{a}^{**}$ does not contain two items from $\{1, 2, 3, 4\}$ with absolute difference ≥ 2 and contains at most one item from $\{4, \dots, n-1\}$.

Case 1.1 $n > 2m + 2$. Assume that there is no item $a \geq 4$. Then there must be at least three items 3. But the triple $(3, 3, 3)$ could be replaced by $(2, 2, 5)$ using (27). Thus there is an item $a \geq 4$ which immediately leads to $2^{m-1}(n-2m+2)^1$.

Case 1.2 $n = 2m + 2$. If there is an item $a \geq 5$ then there is no item 1 and hence $n \geq 2(m-1) + a > 2m + 2$, a contradiction. Thus the only possibility is $2^{m-2}3^2$.

Case 1.3 $n = 2m + 1$. If there is an item $a \geq 4$ then there is no item 1 and hence $n \geq 2(m-1) + a > 2m + 1$, a contradiction. Thus the only possibility is $2^{m-1}3^1$.

Case 2 $n \leq 2m$. If $\mathbf{a}^*$ contains an item $a \geq 3$ then it must also contain an item 1, a contradiction. Thus $\mathbf{a}^*$ contains only items 1 and 2 which yields the assertion. $\square$

7. Proof of [Theorem 3](#)

We have

$$F_{w_3}(a) = \sum_{k=1}^n \left(\binom{n-a}{k} + \binom{n-a}{k-a} \right) \frac{1}{\binom{n}{k}k}.$$

We need a much simpler representation of $F_{w_3}(a)$ and apply the following lemma from [7].

Lemma 11. *If a, b, c are integers such that $a \geq 0, b > 0, c \geq a + b$ then*

$$\sum_{j=0}^a \binom{a}{j} \frac{1}{(b+j)\binom{c}{b+j}} = \frac{1}{b\binom{c-a}{b}}.$$

In fact, we need this lemma only in the case $c = a + b$ and in the case $b = 1$. In these cases the lemma can be proved in a straightforward way.

Lemma 12. *We have*

$$\sum_{k=1}^n \binom{n-a}{k-a} \frac{1}{\binom{n}{k}k} = \frac{1}{a}.$$

Proof. We have

$$\sum_{k=1}^n \binom{n-a}{k-a} \frac{1}{\binom{n}{k}k} = \sum_{j=0}^{n-a} \binom{n-a}{j} \frac{1}{(a+j)\binom{n}{a+j}}$$

and may apply [Lemma 11](#) with $a := n - a, b := a, c := n$. $\square$

Lemma 13. *We have*

$$\sum_{k=1}^n \binom{n-a}{k} \frac{1}{\binom{n}{k}k} = \sum_{k=a+1}^n \frac{1}{k}.$$

Proof. We have

$$\begin{aligned} \sum_{k=1}^n \binom{n-a}{k} \frac{1}{\binom{n}{k}k} &= \sum_{k=1}^n \left(\sum_{l=0}^{n-a-1} \binom{l}{k-1} \right) \frac{1}{\binom{n}{k}k} \\ &= \sum_{l=0}^{n-a-1} \left(\sum_{k=1}^n \binom{l}{k-1} \frac{1}{\binom{n}{k}k} \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{l=0}^{n-a-1} \left(\sum_{j=0}^l \binom{l}{j} \frac{1}{(1+j)\binom{n}{1+j}} \right) \\
&= \sum_{l=0}^{n-a-1} \frac{1}{n-l} \\
&= \sum_{k=a+1}^n \frac{1}{k}.
\end{aligned}$$

Here the second to last identity follows from [Lemma 11](#) with $a := l$, $b := 1$, $c := n$. $\square$

Now we are ready to present the simpler form of $F_{w_3}(a)$ which follows immediately from [Lemmas 12](#) and [13](#).

Theorem 5. *We have*

$$F_{w_3}(a) = \frac{1}{a} + \sum_{k=a+1}^n \frac{1}{k}.$$

Consequently, for the proof of [Theorem 3](#), we work with the function

$$g(a) = 1 + a \sum_{k=a+1}^n \frac{1}{k} \quad (= aF_{w_3}(a)).$$

We have for all $a \in \{2, \dots, n-1\}$

$$g(a-1) + g(a+1) - 2g(a) = (a-1) \left(\frac{1}{a} + \frac{1}{a+1} \right) - \frac{2a}{a+1} = -\frac{1}{a} < 0,$$

and hence we may apply [Theorem 4](#) with $c = n$ and this theorem immediately yields the statements in [Theorem 3](#). $\square$

8. Concluding remarks

We proved [Theorem 2](#) only for $n \geq 11$. For smaller n , some of the used inequalities are not true, which leads to some differences. But for $3 \leq n \leq 10$, the values $g(a)$, $a \in \{1, \dots, n-1\}$, can be concretely determined and optimal partitions can be determined by complete search or using a slight modification of the algorithm from [\[4\]](#). We omit the details.

With analogous methods, the average cardinality of the $\mathcal{A}$ -upper approximation and of the $\mathcal{A}$ -lower approximation, respectively, can be optimized. In the case $j = 2$ (and in a trivial way also for $j = 3$) there are some small differences concerning the used inequalities which leads to some small differences concerning the optimal partitions. We again omit the details.

An inspection of the proof of [Lemma 1](#) and the application of [Lemma 12](#) lead to the identity

$$\sum_{X \subseteq [n]} w_3(|X|) |\mathcal{A}^-(X)| = \sum_{i=1}^m a_i \sum_{k=0}^n \binom{n-a_i}{k-a_i} w_3(k) = \sum_{i=1}^m a_i \frac{1}{a_i} = m.$$

Consequently,

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{A}^-(X)|}{\binom{n}{|X|} |X|} = m. \quad (32)$$

This identity is similar to the well-known AZ-identity [\[1\]](#) (for a textbook see [\[3\]](#)):

Theorem 6 (AZ-identity). *Let $\mathcal{F}$ be a family of nonempty subsets of $[n]$. For $X \subseteq [n]$ let*

$$\mathcal{F}(X) = \bigcap_{A \in \mathcal{F}: A \subseteq X} A$$

(here the intersection is defined to be empty if there is no $A \in \mathcal{F}$ with $A \subseteq X$). Then

$$\sum_{\emptyset \neq X \subseteq [n]} \frac{|\mathcal{F}(X)|}{\binom{n}{|X|} |X|} = 1.$$

Note that [\(32\)](#) follows directly from [Theorem 6](#) by applying it to the families $\mathcal{F}_i = \{A_i\}$ and summing up over $i = 1, \dots, m$.

Acknowledgments

We would like to thank the anonymous referees for valuable comments. In particular we are grateful for the hint that (32) is a consequence of the AZ-identity.

References

- [1] R. Ahlswede, Z. Zhang, An identity in combinatorial extremal theory, *Adv. Math.* 80 (1990) 137–151.
- [2] G.E. Andrews, *The Theory of Partitions*, Cambridge University Press, Cambridge, New York, Melbourne, 1998.
- [3] K. Engel, Sperner theory, in: *Encyclopedia of Mathematics and its Applications*, vol. 65, Cambridge University Press, Cambridge, New York, Melbourne, 1997.
- [4] K. Engel, T. Radzik, J.-C. Schlage-Puchta, Optimal integer partitions, *European J. Combin.* 36 (2014) 425–436.
- [5] Z. Pawlak, Rough sets, *Int. J. Comput. Inf. Sci.* 11 (1982) 341–356.
- [6] Z. Pawlak, *Rough Sets – Theoretical Aspects of Reasoning About Data*, Kluwer academic publishers, Dordrecht, 1991.
- [7] Tran D. Thu, An AZ-style identity and Bollobás deficiency, *J. Combin. Theory Ser. A* 114 (2007) 1504–1514.